Package ‘thresholdanalysis’

May 27, 2026

Title Threshold Effects in Regression Models

Version 1.0.0

Description Provides a unified framework for detecting and modeling threshold effects in logistic regression, linear regression, and Cox proportional hazards models.
The package implements automated threshold identification using restricted cubic splines (RCS) and generalized additive models (GAM) with profile likelihood optimization.
Key features include:
- Confounder-adjusted analyses with 95% confidence intervals for threshold estimates
- Visualization of odds ratios (OR), hazard ratios (HR), and beta coefficients with 95% CI bands
- Excel-ready output of model coefficients and likelihood ratio test results
- Segmented regression modeling via X1/X2 variables for piecewise linear/nonlinear effects
Primary applications encompass dose-response relationship analysis,
exposure threshold determination in clinical research, and epidemiological studies.

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Encoding UTF-8

Roxygen list(markdown = TRUE)

Imports dplyr, ggplot2, survival, openxlsx, broom, lmtest, mgcv,
purrr, readr, readxl, rms, scales, splines

Suggests knitr, rmarkdown

Config/roxygen2/version 8.0.0

NeedsCompilation no

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Contents

cox_threshold *Cox Proportional Hazards Model with Threshold Effect Analysis*

Description

This function performs a threshold effect analysis using Cox proportional hazards models, including restricted cubic spline (RCS) modeling for the focal variable, threshold determination, and model comparison **adjusted for confounders**. HR+95%CI visualization with explicit threshold marking. Outputs include Excel results and analysis objects.

Usage

```
cox_threshold(formula, data, var_name, rcs_nodes = 4)
```

Arguments

`formula`	A formula specifying the model **including both focal variable (var_name) and confounders**. Example: `Surv(time, status) ~ focal_var + confounder1 + confounder2`
`data`	Data frame containing all variables in the formula
`var_name`	Name of the focal variable to analyze for threshold effects
`rcs_nodes`	Number of knots for RCS (default=4) applied to focal variable

Value

List containing:

- cutoff: Optimal threshold value
- mdl0: Original model with all predictors (focal + confounders)
- mdl1: Segmented model (X1/X2 for focal variable + confounders)
- mdl2: Threshold test model (focal variable + X2 + confounders)
- plot: Visualization object of HR curve with threshold marker

linear_threshold *Linear Regression with Threshold Effect Analysis*

Description

This function performs a threshold effect analysis using linear regression models, including restricted cubic spline (RCS) modeling for the focal variable, threshold determination, and model comparison **adjusted for confounders**. Regression coefficient+95%CI visualization with explicit threshold marking. Outputs include Excel results and analysis objects.

Usage

```
linear_threshold(formula, data, var_name, rcs_nodes = 4)
```

Arguments

`formula`	A formula specifying the model **including both focal variable (var_name) and confounders**. Example: `outcome ~ focal_var + confounder1 + confounder2`
`data`	Data frame containing all variables in the formula
`var_name`	Name of the focal variable to analyze for threshold effects
`rcs_nodes`	Number of knots for RCS (default=4) applied to focal variable

Value

List containing:

- cutoff: Optimal threshold value
- mdl0: Original model with all predictors
- mdl1: Segmented model (X1/X2 for focal variable + confounders)
- mdl2: Threshold test model (focal variable + X2 + confounders)
- plot: Visualization object of coefficient curve with threshold marker

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`logistic_threshold` *Logistic Regression with Threshold Effect Analysis*

Description

This function performs a threshold effect analysis using logistic regression, including restricted cubic spline (RCS) modeling for the focal variable, threshold determination, and model comparison **adjusted for confounders**. OR+95%CI visualization with explicit threshold marking. Outputs include Excel results and analysis objects.

Usage

```
logistic_threshold(formula, data, var_name, rcs_nodes = 4)
```

Arguments

`formula`	A formula specifying the model **including both focal variable (var_name) and confounders**. Example: `outcome ~ focal_var + confounder1 + confounder2`
`data`	Data frame containing all variables in the formula
`var_name`	Name of the focal variable to analyze for threshold effects
`rcs_nodes`	Number of knots for RCS (default=4) applied to focal variable

Value

List containing:

- cutoff: Optimal threshold value
- mdl0: Original model with all predictors (focal + confounders)
- mdl1: Segmented model (X1/X2 for focal variable + confounders)
- mdl2: Threshold test model (focal variable + X2 + confounders)
- plot: Visualization object of OR curve with threshold marker